

Cloning Windows

Tools such as Symantec Ghost (www.symantec.com) clone entire partitions and even hard disks, but these programs cost. You can do the same without installing any additional software. Basically, you need to copy the files to a different location and copy them back when necessary. For fast and easy copying, you should have a hard disk with at least two partitions and a friend who will let you use his computer when your Windows crashes.

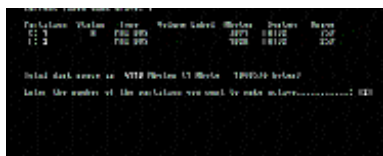
While in Windows, close all open programs, open Command Prompt and switch to the folder where you want to make the copy. Here, type the following command:

```
xcopy c:\*.* /e /c /r /h /y /k
```

This will copy all the files and folders from the C drive to the backup location. By doing it from Windows, the copy operation itself is faster than copying while in DOS and, more importantly, all long filenames are preserved. Using XCopy allows you to copy all folders and subfolders, maintain file attributes and ignore errors (copying using Windows Explorer will throw up an error if it tries to copy the swap file and it will stop copying).

To restore this image, you will have to be able to access the hard disk from Windows. Thus, if you don't have Windows installed on another hard disk or a different partition, you'll need to attach the hard disk to another machine. Run the same procedure as above to copy the files.

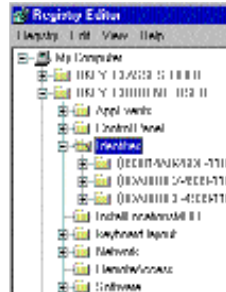
If you do not have access to another machine, you can still have a working backup. Simply dedicate a partition to hold this backup, so that you copy



Use Fdisk to switch from a corrupt installation to a working backup

the files to the root of that partition. You can boot to DOS and use FDisk to switch the active partition. When in trouble, just boot off that partition instead! Thus, you will never have to reinstall Windows.

by saving the HKEY_CURRENT_USER\Software\Microsoft\Office\Outlook\OMI Account Manager\Accounts branch from the Registry. All mail is stored in PST files. As with other Office programs, you can use the Save My Settings Wizard in Outlook 2000 and XP, though it may not back up all your settings.



Outlook Express stores all information in the Windows Registry

C:\Windows\Application Data\Identities. Simply copy the folders to another location; they can later be imported into Outlook Express. The Address Book is stored in a WAB file, typically in C:\Windows\Application Data\Microsoft\Address Book. All

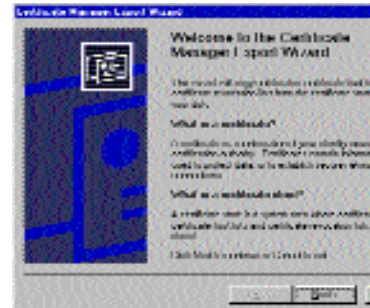
There are fewer things you need to back up in Outlook Express. All mail is stored in DBX files (one file for each mail folder) in a separate folder for each identity. On Windows 9x you will be able to locate the folders for all identities in



Windows is not as simple to restore to the original look and feel as applications, especially if you tweak it quite a bit. Windows stores some settings in the Win.ini and System.ini files in the Windows folder, while many others are distributed across the Registry.

If you modify Windows settings directly from the Registry itself, you should export these registry keys to a folder as you modify them—to apply the same tweaks, simply double-click each file. Another simple way is to use a third-party tool such as Tweak UI, which makes standard tweaks a breeze. Thus, when you reinstall Windows, you can change all those settings in just a matter of minutes.

The fastest and most effective way to restore Windows to its original state is to maintain backups of the entire Registry—this will not only save Windows settings, but also settings of all programs that use the Registry to store their settings. It is extremely vital that the backup be taken when the system is functioning perfectly. After a fresh installation of Windows, after all your most-used applications are



The Certificate Manager Export Wizard takes care of all your digital certificates

account settings and rules are stored with the identity information in the Windows Registry, under the HKEY_CURRENT_USER\Identities key. Simply export this branch and later restore it and Outlook Express will start functioning just as before.

If you use stationery in your e-mail, you will find the files in \Program Files\Common Files\Microsoft Stationery. If you have digital signatures, open **Tools > Options > Security > Digital Signatures**, select each digital certificate and export it to a file. Remember to export the private key too, along with the public key. You can then import them back from the same window.

OS Set-

Windows is installed, you should take the first backup. Next, go on to changing system settings and customising applications and another backup. You should update your backup every time you install a program or change some settings or upgrade software—changes that you know you will want to restore. When you decide to restore the settings after reinstalling Windows, first reinstall all the programs that were installed when the backup was taken, then reboot to DOS. You can then replace the Registry with the one you saved in your backup using the /c switch with regedit. Thus, if you have a file named backup09.reg stored in D:\Backups, you would use the following command:

```
regedit /c d:\backups\backup09.reg
```

Windows 98 includes a tool called Registry Checker that, by default, takes daily backups of the Registry. This keeps only the last five backups by default, but it also maintains one backup of the Windows installation. You can restore the system of the backups by running scanreg /r from DOS. Along with the registry utility also saves and restores the Windows